Author Response to Reviewer 1P2E (Rating: 4, Borderline Accept)

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We thank Reviewer 1P2E for the thoughtful and constructive feedback. Below we address each point in detail. All changes in the revised manuscript are marked in blue text; line references (e.g., `[L42--44]`) refer to the revised `main.tex`.

R1.1: Supplement empirical findings with a theoretical discussion explaining why simple covariance invariants outperform more complex representations (bias-variance trade-off, parameterization complexity, dataset size). **Response:** We have substantially expanded the "simplicity principle" discussion in Section 6 `[L604--605]`. The revised text `[L605]` now provides an explicit parameter-count analysis: the 5 scalar invariants operate at $\sim$180 samples per covariance parameter on ESOL ($N_{\text{train}} \approx 900$), whereas the lowrank variant (k=128) operates at $\sim$7—well below the threshold for reliable estimation. We connect this to classical model selection theory, showing that the bias-variance tradeoff penalizes higher-dimensional representations at the sample sizes typical of molecular property datasets (642–4,200 molecules). The new feature dimension ablation (Table 6, Section 5.6, `[L484--508]`) provides independent confirmation: even for the mean $\boldsymbol{\mu}$, RMSE monotonically degrades from D=256 to D=2,048, demonstrating that this is a fundamental statistical constraint rather than an engineering limitation.

Specific changes:

- Bias-variance theoretical analysis expanded: `[L605]`
- Feature dimension ablation section added: `[L484--508]`
- Feature dimension ablation table (Table 6): `[L489--504]`
- Conclusion updated with bias-variance finding: `[L618]`

R1.2: Conduct supplementary experiments on classification benchmarks. **Response:** We have added classification experiments on BACE (1,513 molecules, β-secretase inhibition) and BBBP (2,038 molecules, blood-brain barrier permeability) in a new Section 5.5 `[L461--481]`. Results (Table 7, `[L466--479]`) confirm the property taxonomy extends to classification: conformer features hurt BACE by −4.4% AUROC and show no benefit on BBBP (+0.1%). This is consistent with the taxonomy—β-secretase binding is an electronic/binding property, and BBB permeability is governed primarily by topological features. Per-seed results are provided in Appendix K `[L1041--1064]`.

Specific changes:

- New classification benchmark section: `[L461--481]`
- Classification results table (Table 7): `[L466--479]`
- Scope statement updated in Introduction: `[L151]`

- Abstract updated to include classification: [L80]
- Per-seed classification appendix: [L1041--1064]

R1.3: Validate key findings on larger or industrially sourced datasets. **Response:** We acknowledge this as an important limitation in the revised Conclusion [L620]. Our benchmarks span 642 to 133K molecules across academic datasets (MoleculeNet, QM9, MARCEL); validation on larger industrially sourced datasets (ChEMBL, ZINC, proprietary ADMET panels) would strengthen the taxonomy's generalizability. We note this as a priority for future work and have added explicit language to this effect.

Specific changes:

- Limitations paragraph updated: [L620]

R1.4: Increase figure resolution, standardize metrics, correct typographical errors. **Response:** We have conducted a full proofreading pass. All figures are exported at ≥300 DPI. Evaluation metrics are standardized (RMSE throughout for regression, AUROC for classification, [L461--481]). Typographical errors and formatting inconsistencies have been corrected across tables and main text (see blue-marked changes throughout).

Specific changes:

- Figures re-exported at ≥300 DPI: all `figures/*.pdf` files
- Metrics standardized: RMSE for regression, AUROC for classification [L461--481]
- Typographical corrections throughout (see blue-marked text)

R1.5: Provide clearer descriptions of scaffold split generation and number of replicates. **Response:** We have expanded Section 3.7 (Training Details, [L274]) with explicit descriptions of the Murcko scaffold split procedure (Bemis–Murcko generic scaffolds extracted via RDKit, sorted by frequency, assigned to splits maintaining 80/10/10 ratios) and seed counts (3 seeds for the main benchmark, 10 seeds for statistical validation experiments).

Specific changes:

- Scaffold split procedure details: [L274]
- Replicate counts explicitly stated: [L274]
- Stratified splits for classification explained: [L464]

Summary of changes relevant to Reviewer 1P2E:

- Added classification benchmark on BACE and BBBP [L461--481], Appendix K [L1041--1064]
- Added feature dimension ablation $D \in \{256, 512, 1024, 2048\}$ [L484--508]
- Expanded bias-variance theoretical analysis [L604--605]
- Clarified scaffold split procedure and replicate counts [L274]
- Added larger-dataset limitation acknowledgment [L620]

- All figures at $\geq$300 DPI; metrics standardized; typos corrected
- All changes marked in blue text